

WASP-52b

R-0342

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_221011 · created 2026-07-18T07:20:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459863.6817 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.013
Transit depth (Rp/Rs)^2	2.27 +/- 0.4 [%]
Orbital Inclination (inc)	85.69 +/- 1.02
Airmass coefficient 1 (a1)	1.105 +/- 0.014
Airmass coefficient 2 (a2)	-0.08 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	None
Optimal Aperture	3.58
Optimal Annulus	5.23
Transit Duration (day)	0.0755 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.013 vs 0.1646 (deviation 1.04σ)
Duration fit vs published	0.0755 d vs 0.0758 d (deviation 0.05σ)
Mid-time O–C	13.2 min
Depth SNR	11.5
Residual scatter	1.29 %

Light curve

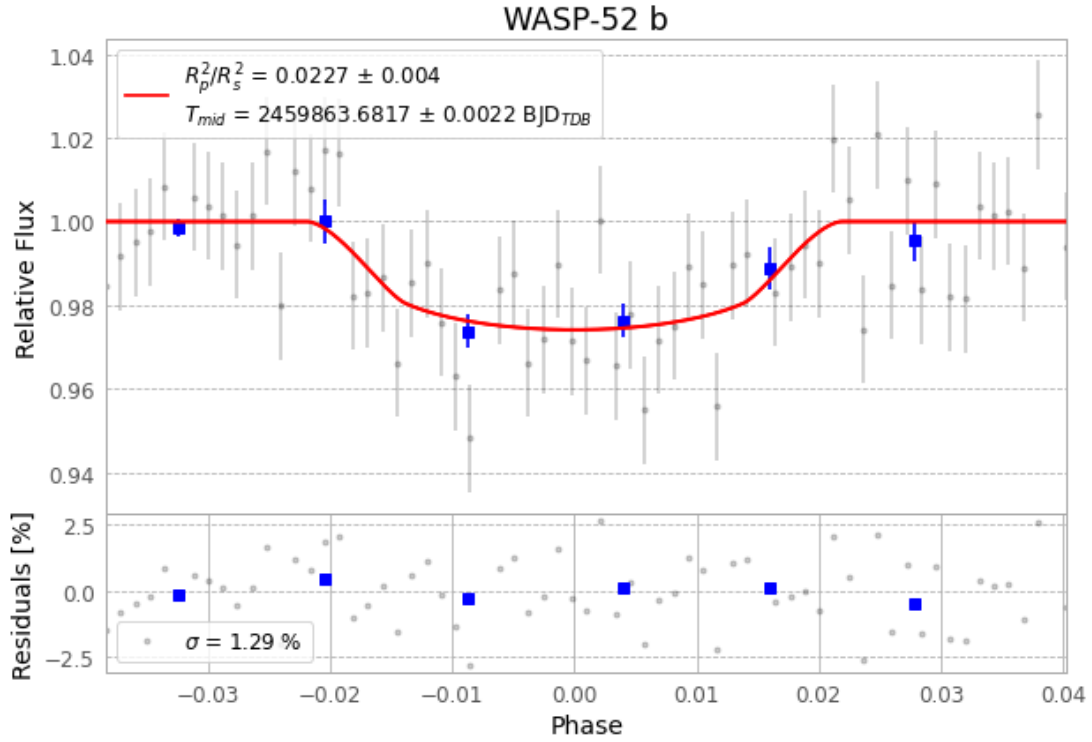


Figure 1 — FinalLightCurve_WASP-52 b_2022-10-10.png (SHA-256 c3121a1e28bb648b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [446, 299], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 769a9802e42870e7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

66 FITS frames (43 MB), WASP-52221011022716.FITS → WASP-52221011055415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221011.log (dd0a758c245d...), FinalLightCurve_WASP-52 b_2022-10-10.png (c3121a1e28bb...), FinalLightCurve_WASP-52 b_2022-10-10.csv (69213f1ee7ac...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 07:21Z.